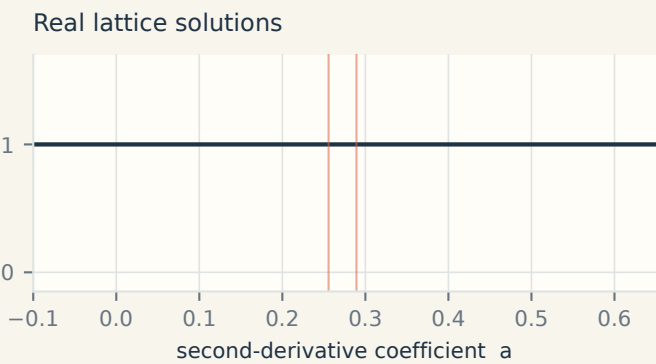
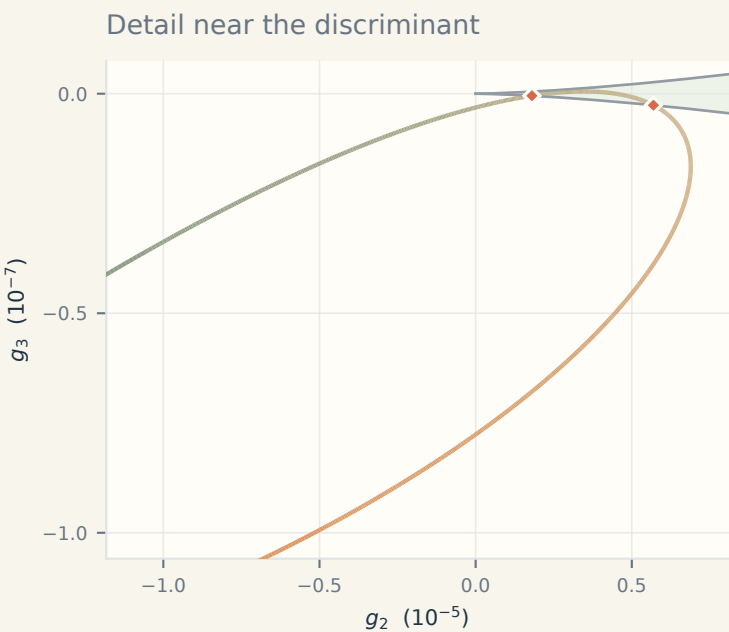
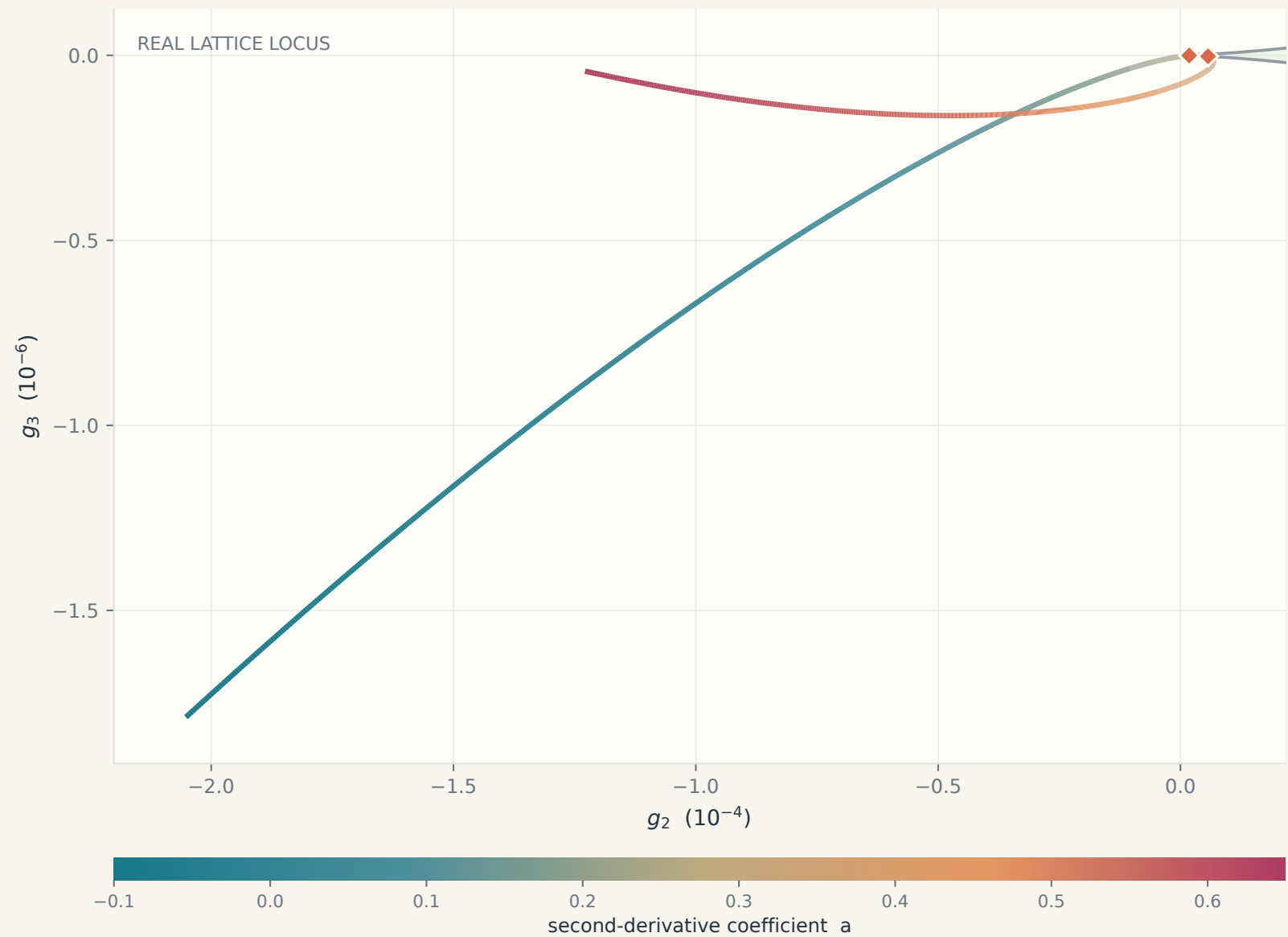


Kawahara-KS · the dissipative curve

$(n, p) = (2, 4) \cdot q = 4$

$$u'''' + u''' + au'' + a_1(a)u' - 840u^2 + C(a) = 0$$



Grey cusp: $\Delta = 0$
Pale region: $\Delta > 0$
Diamonds: real degeneration parameters

The odd derivative removes the two-wave freedom: one wave per compatible fixed equation.
The locus resolves several different passages through the discriminant.
Lattice of the full twisted period group Γ · Real invariants do not alone imply a bounded real profile.